

CONTENTS OF WORKSHOP

- Installation of Arduino IDE
- Importing data and plotting waveforms
- Interfacing of Arduino with sensor, Bluetooth, mobile phone and LCD display for IoT applications
- Design and implementation of moving-average, HPF, LPF, BPF and BSF (Butterworth and Chebyshev type) in Arduino
- Implementation of auto-correlation function and FFT
- Development of Automatic Pulse rate measurement system

FEE PAYMENT

Payment link:

<https://www.onlinesbi.sbi/sbicollect/collethome.htm>

Steps for payment:

1. Select "Educational Institutions"
2. Search for Payee "IIIT Kottayam" and click on it
3. Payment Category – Select "Arduino Workshop"
4. Enter the details and make payment

ABOUT THE PROGRAM

The workshop focuses on programming Arduino micro-controllers to develop Internet of Things (IoT) applications.

Micro-processors, micro-controllers and real-time embedded systems will be introduced. Next, evolution of micro-controllers elucidating their significance at various stages will be briefed. Also, a special note on ARM micro-controllers describing its architecture, instruction set and basic programming will be discussed.

This workshop enables the participants to gain knowledge of interfacing hardware components with Arduino IDE through hands-on sessions. Further, the digital signal processing techniques like filtering, auto correlation function and Fast Fourier Transform (FFT) using Arduino micro-controller will be implemented. The participants will be able to develop Arduino-based IoT applications after the hands-on training sessions.

Students, scholars and faculty from all disciplines of Engineering and Science who wish to explore the realm of Arduino based IoT applications can participate in this workshop.



Indian Institute of
Information Technology
Kottayam



**Department of ECE Jointly
with I²CS Research Group
presents**

**2-Day Hands-on Workshop
on**

**Arduino Programming for
IoT Applications**

June 17 - 18, 2023



<https://www.iiitkottayam.ac.in>

<http://i2cs.iiitkottayam.ac.in/>

ABOUT THE INSTITUTE

Indian Institute of Information Technology, Kottayam is one among the IIITs that have been established as "Institutions of National Importance" by Ministry of Education, Govt. of India under the ambit of IIIT (PPP) Act 2017. The institute offers 4 year B.Tech and B.Tech(Hons) program with specialized courses covering all major research domains of Computer Science and Electronics & Communication Engineering. The students are admitted against the JoSAA/CAB counselling against the JEE main rank list. Institute also offers MTech in CSE and PhD program in CSE, ECE and Mathematics.

ABOUT DEPARTMENT OF ECE

Department of ECE offers both B.Tech and B.Tech(Hons) degree, with courses covering wide domain of Electronics Circuits, VLSI & Embedded Systems, Signal Processing, Communication Systems, Antenna Design, Microwave Engineering, Instrumentation, Artificial Intelligence and Machine learning. The department also offers PhD program in these domains. The department has a vibrant faculty community who are involved in the cutting edge research activities through I²CS research group.

REGISTRATION

<https://shorturl.at/kaORX>

Registration Fee :

UG/PG students : **Rs. 600 /-**
Ph.D scholars : **Rs. 800 /-**
Faculty Members : **Rs. 1000 /-**



Accommodation will be provided to the registered participants in the campus, on payment basis.

PROGRAM CONVENER

- **Dr. Kala S**
Assistant Professor, I²CS Lead
HOD - Department of ECE
IIIT Kottayam

RESOURCE PERSONS

- **Dr. S. Venkatesh**
Assistant Professor
Department of ECE
IIIT Kottayam
- **Dr. G. Narendra Kumar Reddy**
Assistant Professor
Department of ECE
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ORGANIZING TEAM

- **Dr. Ananth A**
- **Dr. Santhos Kumar**
- **Dr. S. Venkatesh**
- **Dr. Milind Thomas**
- **Dr. Minu A Pillai**
- **Dr. Della Thomas**
- **Dr. Rajesh G**
- **Mr. K. Vishal**

PATRONS

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PROGRAM CO-ORDINATOR

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